

Your grade: 100% - Your highest score is 100%. You have earned 100% of the available points.

Graded Assignment: Analyzing Medical Volumes

Next item →

- Which option below is not a valid way to load images? 1 / 1 point
 - ☒ To resample a medical volume to a new resolution so you can compare two volumes.
 - ☐ To segment certain parts of the body or organs for further analysis.
 - ☐ To create a labeled dataset for training a deep learning model.
 - ☐ To highlight specific regions of interest for visualization.

Correct

You cannot resample an image directly in the app. This is better done using code as shown in Module 2 of this course.

Due Aug 5, 11:59 PM EST Submitted Aug 5, 11:59 AM EDT	Attempts Unlimited	Try again
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- True or False: Labeling medical images usually requires you to manually label some pixels/voxels. 1 / 1 point
 - ☐ True
 - ☐ False

Your grade
To pass you need at least 70%. We keep your highest score.

View submission **See feedback**

☒ **Correct**

While the app includes a tool to help speed the labeling process, you will likely need to do some manual labeling.
- How many different labels can you create in the Medical Image Labeler App to label part of the anatomy? 1 / 1 point
 - ☐ 1
 - ☐ 2
 - ☐ 10
 - ☒ As many as required.

☒ **Correct**

Instructions for remaining problems

Use the `AnalyzingLabeledVolumes_mlx` file provided with the course files to answer the following questions. You'll work with a labeled volume, `T1_3T_2018.nii`, and the corresponding `T1_3T_2018` MRI data provided in the course Data folder.

- The labeled volume included in the course files using the `medicalVolume` function. How many unique values are there in the `Voxels` property? 1 / 1 point

3

☒ **Correct**

The 3 unique values are 0, 1, and 2 corresponding to the background, left hemisphere and right hemisphere, respectively.

Labels for the left and right hemispheres were created by the user in the Medical Image Labeler App while voxels that are 0 were unlabeled in the app.
- What is the volume of the left hemisphere in voxels? 1 / 1 point
 - ☒ 28849
 - ☐ 29365
 - ☐ 6.1e5
 - ☐ 6.2e5

☒ **Correct**

Correct. The first label is the left hemisphere and the volume is 28849 voxels.
- True or False 1 / 1 point

Labels generated by a deep learning model do not need to be inspected.

 - ☐ False
 - ☐ True

☒ **Correct**

Knowing the model details and formatting the input according to the model requirements can help improve the accuracy of your results, but in the end, they are just predictions. You should always inspect the results.